

MaternalCare OB-GYN Clinical System

End-to-End Features, Architecture & Clinical Workflow Specification

Integrated Obstetrics & Gynecology Patient Records, Scheduling & Smart Rx Prescription Platform

Version 2.0 | Production Release | Live URL: <https://medical-care-system-one.vercel.app/>

1. EXECUTIVE OVERVIEW & CORE VALUE PROPOSITION

■ Zero-Lockin Dual Excel & Cloud Architecture

- Combines instantaneous client-side responsiveness with live multi-tab Excel database synchronisation (OBGYN_Clinic_Database.xlsx).
- Supports Chrome File System Access API for transparent local disk auto-saving without requiring recurring file dialogs.
- Provides 1-click Excel Export, Import, and Factory Default Database Restoration across all clinic workstations.

■ Role-Tailored Dual Clinical Portals

- Lead Doctor (MD / FPOGS): Full clinical authorization (Gestational metrics, Prescriptions, High-Risk Watchlist, Record Deletion).
- Assistant Nurse (RN / Triage): Triage vitals registration, patient intake, appointment scheduling, and WhatsApp/SMS reminder dispatching.
- Fast Security PIN Authentication: Pre-configured demo PINs and 4-digit practitioner credentials stored securely in Excel database.

2. PRACTITIONER AUTHENTICATION & STAFF MANAGEMENT

■ Secure Portal Login & Fast Profile Switcher

- Visual Doctor vs Assistant Nurse toggle tabs with personalized practitioner profile selectors.
- 4-digit security PIN verification with intuitive default helper pills (Dr. Sarah Jenkins MD: 1234, Nurse Maria Santos RN: 0000).
- Instant staff registration directly from the login portal (+ Register a Nurse or New Doctor) without administrative lockout.

■ Role Permission & Security Matrix

- Doctor-Exclusive Privileges: Modifying Last Menstrual Period (LMP), Gravida/Para parity, Issuing & Signing Prescriptions (Rx Pad), Deleting Checkup or Patient Records.
- Nurse-Safe Boundaries: Protected fields with Doctor Authorization badges to prevent accidental clinical alterations while enabling fast intake and vital triage logging.

3. INTELLIGENT OBSTETRICS ENGINE & GESTATIONAL CALCULATIONS

■ Automated Naegele's Rule & Biometric Calculations

- Estimated Date of Delivery (EDD): Automatically derived using Naegele's Rule (LMP + 7 days - 3 months + 1 year).
- Age of Gestation (AOG): Calculated to the exact day (Weeks + Days) in real-time against current date.
- Estimated Conception Date: Accurately calculated as LMP + 14 days for comprehensive patient timeline counseling.
- Gestational Trimester Cohort: Categorized dynamically into 1st Trimester ($\leq 13w\ 6d$), 2nd Trimester (14w - 27w 6d), 3rd Trimester ($\geq 28w$), and Post-Term ($\geq 40w$).

■ High-Risk & Delivery Watchlist

- Identifies and escalates patients approaching term (≥ 32 weeks AOG or within 30 to 45 days of expected delivery).
- Interactive cohort distribution charts showing active clinic patient census broken down by pregnancy trimester.

4. INTELLIGENT SCHEDULING & INTERACTIVE CALENDAR SYSTEM

■ Nurse-Centric Interactive Monthly Calendar

- Full-month calendar grid with previous/next month navigation, "Today" quick jump, and single-day filtering.
- Color-Coded Status Badges: Completed consultations are tagged in Green (Done), while upcoming bookings are tagged in Orange (Scheduled).
- Collapsible Upcoming Queue: [Hide List] / [Show List] toggle that seamlessly expands the calendar to 100% full-screen width.

■ Smart Slot Booking & Conflict Prevention

- Prevents double-booking: Automatically scans booked consultations for the selected date and disables taken time slots with (Taken / Booked).
- Pre-fills earliest available consultation window and provides instant modal feedback.

■ 1-Click WhatsApp & SMS Consultation Reminders

- Generates pre-formatted bilingual clinical reminder templates containing patient name, scheduled date, time slot, and clinic location.
- Dispatches directly to patient's registered mobile phone via native WhatsApp or SMS with a single click.

5. PATIENT DIRECTORY & ADAPTIVE TWO-COLUMN MEDICAL CHART

■ Desktop Collapsible Sidebar vs Mobile Adaptive Layout

- Computer / Desktop View: [Hide List] / [Show Patients] toggle collapses the patient directory sidebar, giving the clinical chart 100% full-screen width for spacious recording.
- Mobile View: Optimized single-screen navigation with sticky back button and responsive touch cards.
- Multi-Filter Directory Search: Instant real-time filtering by Patient Name, Phone Number, Home Address, Clinical Status (Active/Inactive), and Gestational Trimester.

■ Comprehensive Patient Demographic & History Profile

- Full obstetric history (GxPy), Patient Age, Blood Type, Contact Details, and Home Address.
- Illness & Medical History Banner: Dedicated collapsible card for documenting Pre-existing Conditions, Surgical History, Family History, and Drug Allergies with high-visibility warning badges.
- Emergency Contact Management: Records Next of Kin name, relationship, emergency phone number, and physical address.
- Protected Clinical Status: Read-only Active Prenatal badge in chart header, strictly editable via authorized "Edit Info" modal.

6. CONSULTATIONS, VITALS TRIAGE & CLINICAL CHECK-UP TABLE

■ Dual Doctor-Nurse Inline Row Entry

- Top input row for instant recording of Consultation Date, Vitals (Weight in kg, Blood Pressure BP, Fetal Heart Rate FHR in bpm), Clinical Diagnosis, Procedure / Treatment, and Follow-Up Date.
- Interactive Date Picker: Allows logging visits for current date or backdating past consultations seamlessly.
- Automatic Consultation Completion: Saving a checkup automatically tags any pending scheduled appointment for that patient as "Done".

■ Historical Consultation Timeline & Medication Badges

- Chronological visit audit trail displaying vitals, findings, and follow-up schedules.
- Live Prescription Badges: Displays Prescribed Medications (X): Generic Name (Dosage) directly in the checkup row with 1-click Rx re-print access.

7. DOCTOR'S OFFICIAL MEDICAL PRESCRIPTION PAD (RX)

■ Professional A4 / US Letter Rx Pad Generation

- Clean, standardized medical layout complete with Clinic Header, Patient Info, AOG / EDD, Allergies Alert, Rx Symbol, Signatures, and PRC / PTR / S2 License Credentials.
- Digital Signature & Official Medical Stamp options ready for physical printing or PDF export.

■ Automatic Prescription Archiving upon Print & Save

- Zero Hardcoded Defaults: Starts with a clean, blank medication list (0 meds) or loads previously saved medicines.
- Auto-Save on Print: Clicking [Print Prescription] or [Save to Checkup Record] automatically writes all prescribed medications permanently into the patient's clinical checkup record and Excel database.
- 1-Click "Clear All" action and fast 1-click OB-GYN preset formulations (Prenatal Multivitamins, Iron + Folic Acid, Calcium + D3, Progesterone / Duphaston, Isoxsuprine, Paracetamol, etc.).

8. TECHNICAL ARCHITECTURE, SECURITY & OFFLINE CAPABILITIES

■ Frontend Stack & Styling System

- Core Framework: React 19 + TypeScript + Vite for instant sub-second bundling and zero-runtime latency.
- TailwindCSS Design System: Modern healthcare color tokens, dark glassmorphism, responsive grid layouts, and high-contrast typography.
- Lucide Healthcare Icons: Crisp vector icons for triage vitals, fetal metrics, prescriptions, and calendar operations.

■ Multi-Tab Excel Database Schema (OBGYN_Clinic_Database.xlsx)

- Sheet 1: Patients Directory - ID, Full Name, Age, Phone, Address, Blood Type, Gravida, Para, LMP, Illness History, Allergies, Status, Emergency Contact.
- Sheet 2: Clinical Checkups - ID, Patient ID, Visit Date, BP, Weight, FHR, Diagnosis, Procedure, Follow-Up Date, Prescriptions JSON, Notes.
- Sheet 3: Appointment Schedule - ID, Patient ID, Patient Name, Date, Time Slot, Reason, Status (Scheduled / Completed / Cancelled), Reminded.
- Sheet 4: Practitioner Accounts - ID, Full Name, Title, Role (DOCTOR / NURSE), PIN Code, Avatar.

MaternalCare OB-GYN System - Built for Clinical Excellence & Patient Safety

All features are fully deployed, verified with strict TypeScript builds, and live in production.